

Corrected q=5,6,7 feature contraction and common-noise anisotropy tail

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-08-02 · this version issued 2026-08-02 · v1.0

1. Purpose and scope

R-142 found the correct two-feature owner (U, Φ) and a signed rational collar band, but its displayed three-layer $q = 5, 6, 7$ coherence witness did not obey its own shell convention. R-143 repairs that finite geometry and then changes the proof coordinate. Instead of estimating the adverse trace feature term by term, it places every already registered favorable square in one feature $\mathcal{Y}$ and asks for one contraction from $\mathcal{Y}$ to $\mathcal{U}$. This is an exact finite-set Douglas criterion, not another absolute-value budget.

The durable conclusions are:

1. the old integer collision is a real $q = 6, 7, 8$ collision at physical output 4229940, not the declared $q = 5, 6, 7$ collision at 2114970;
2. an exact dyadic same-sign $q = 5, 6, 7$ family is supplied, with canonical harmonics $(10, 20, 40)$ and carriers $(4M, 2M, M)$;
3. the complete source and sextic reserves enter one positive feature exactly once, and finite-set positivity is equivalent to a contraction or, equivalently, a mixed Gram inequality;
4. the raw R-141 signature and the action have opposite signs and a factor $1/2$; this conversion is now an executable firewall;
5. the A1 family-lock covariance has an exact scalar-plus-anisotropic common-probe realization, and its high-shell derivative trace has an explicit summable lattice bound;
6. the stronger mixed-Gram route still cannot be evaluated on production data because the temporally faithful U/Φ jets, cross reveals, source-null coordinates, and returned-low blocks are not registered.

Every statement is finite-cutoff and positive-floor unless an infinite lattice upper bound is explicitly labelled conditional. The finite Gram theorem is exact on any supplied feature family; it does not assert that the missing production family passes it. The $N = 27$ and $N = 49$ tail comparisons use previously registered illustrative headrooms, not production margins. No A13 gate, matching identity, absolute anchor, Nelson estimate, removal, interacting measure, Sector-A closure, or tier promotion follows.

2. Audit and correction of the R-142 collision

R-142 uses the sharp shell convention

$$2^{j-1} < N \leq 2^j \tag{2.1}$$

for a root carrier N , and the moved rational harmonic n lies at gap q when

$$2^{q-2} < n \leq 2^{q-1}. \tag{2.2}$$

The physical output of $\cos(2nNx)$ is $K = 2nN$.

The old data were

$$n = (17, 33, 65), \quad N = (124410, 64090, 32538). \quad (2.3)$$

They satisfy $nN = 2114970$, so they do form an integer collision. But the physical output and sharp shell data are

$$\begin{aligned} K &= 2nN = 4229940, \\ j &= (17, 16, 15), \quad m = 23 \end{aligned} \quad (2.4)$$

and hence

$$m - j = (6, 7, 8). \quad (2.5)$$

Equivalently, (2.2) assigns $(17, 33, 65)$ to $(6, 7, 8)$. Thus the collision survives only as a non-dyadic $q = 6, 7, 8$ witness. The R-142 coefficient-band sign theorem is independent of this example and remains intact.

Now let $M = 2^J$ and choose any even $t \in \{10, 12, 14, 16\}$. Put

$$(n_5, n_6, n_7) = (t, 2t, 4t), \quad (N_5, N_6, N_7) = (4M, 2M, M). \quad (2.6)$$

Then

$$2n_5N_5 = 2n_6N_6 = 2n_7N_7 = 8tM. \quad (2.7)$$

For the canonical $t = 10$ choice the output is $80M$. Its output shell is $J + 7$, while the root shells are $(J + 2, J + 1, J)$, so the gaps are exactly $(5, 6, 7)$. Moreover the registered sign formula

$$(-1)^{n+1}g_n > 0 \quad (2.8)$$

gives

$$g_{10} < 0, \quad g_{20} < 0, \quad g_{40} < 0. \quad (2.9)$$

Thus the three contributions are genuinely same-sign at one physical output, with no phase adjustment. The primary certificate also recomputes the coefficients from the A1 inputs at $\delta = 0.2$ and test floor $1/9$.

There is a second legal family with $t = 9$. Its signs are mixed, but a phase choice can align them. It is useful for adversarial search, whereas the even family (2.6) is the cleaner sign-regression fixture. Neither family is yet a production counterdirection because the missing owner blocks can change the full action sign.

3. Raw signature versus action: sign and factor firewall

After R-142 innovation compression, write

$$F = (U, \Phi), \quad S_2 = \text{diag}(+I, -I), \quad K(F) = \langle F, S_2 F \rangle = \|U\|^2 - \|\Phi\|^2. \quad (3.1)$$

The compressed action owner is not K but

$$\boxed{\mathcal{P}_{\text{comp}}(F) = -\frac{1}{2}K(F) = \frac{1}{2}\|\Phi\|^2 - \frac{1}{2}\|U\|^2.} \quad (3.2)$$

Consequently

$$D^2\mathcal{P}_{\text{comp}} = -\frac{1}{2}D^2K. \quad (3.3)$$

A positive raw-signature eigenvalue is action-adverse; a negative eigenvalue of an already action-signed Hessian is adverse. The certificates check (3.3) on an exact polynomial fixture. This rule must be attached to every future matrix output so that a solver cannot silently reverse the verdict.

4. The complete one-use positive feature

Fix one finite cutoff, positive floor, temporally faithful chart π , and recombine every complete owner before taking direct sums. Let $\Phi_\pi(h)$ be R-142's conditional mean-current feature and let $\mathcal{U}_\pi(h)$ be its actual shared-probe trace feature. Define

$$\boxed{\mathcal{Y}_\pi(h) = \Phi_\pi(h) \oplus \sqrt{\frac{9}{10}} h \oplus \sqrt{\frac{3}{10}} |Z_{K,h}|^2 Z_{K,h}.} \quad (4.1)$$

Expectations, spatial integration, complete-owner sums, and auxiliary probes are part of the Hilbert norms. The complete R-141/R-142 core becomes

$$\boxed{2\mathcal{A}_\pi(h) = \|\mathcal{Y}_\pi(h)\|^2 - \|\mathcal{U}_\pi(h)\|^2.} \quad (4.2)$$

Indeed,

$$\frac{1}{2} \left\| \sqrt{\frac{9}{10}} h \right\|^2 = \frac{9}{20} \|h\|^2, \quad \frac{1}{2} \left\| \sqrt{\frac{3}{10}} |Z|^2 Z \right\|^2 = \frac{3}{20} \mathbb{E} \|Z\|_6^6. \quad (4.3)$$

Thus source and sextic are present exactly once. The R-125 forest is an alternate coordinate for the already recombined variance owner and is not a fourth reserve. Predictable mean and returned low must remain inside the complete Φ /low assembly rather than being appended again.

For a relative endpoint, centering alone is not harmless:

$$\begin{aligned} 2[\mathcal{A}(h) - \mathcal{A}(h_0)] &= \|\Delta Y\|^2 - \|\Delta U\|^2 \\ &\quad + 2\Re\langle Y_0, \Delta Y \rangle - 2\Re\langle U_0, \Delta U \rangle. \end{aligned} \quad (4.4)$$

The baseline cross terms belong to the exact low/Feshbach block. They may not be deleted unless matching or an absolute anchor has first been proved.

5. Finite-set Douglas theorem

Let $h_1, \dots, h_N$ be any finite admissible endpoint family and set

$$(G_Y)_{ab} = \langle \mathcal{Y}(h_a), \mathcal{Y}(h_b) \rangle, \quad (G_U)_{ab} = \langle \mathcal{U}(h_a), \mathcal{U}(h_b) \rangle. \quad (5.1)$$

Theorem 5.1. The following are equivalent:

1. $G_U \preceq G_Y$;
2. there is a contraction $C : \overline{\text{span}}\{\mathcal{Y}(h_a)\} \rightarrow \mathcal{H}_U$ with $C\mathcal{Y}(h_a) = \mathcal{U}(h_a)$ for every a ;
3. for every coefficient vector c ,

$$\left\| \sum_a c_a \mathcal{U}(h_a) \right\|^2 \leq \left\| \sum_a c_a \mathcal{Y}(h_a) \right\|^2. \quad (5.2)$$

Proof. Write $Yc = \sum_a c_a \mathcal{Y}(h_a)$ and $Uc = \sum_a c_a \mathcal{U}(h_a)$. The Gram inequality is exactly $\|Uc\| \leq \|Yc\|$. Hence $Yc = 0$ implies $Uc = 0$, so $C_0(Yc) = Uc$ is well-defined and contractive on $\text{Ran } Y$ and extends to its closure. Conversely, $U = CY$ and $\|C\| \leq 1$ imply $U^*U \preceq Y^*Y$. $\square$

The optimal finite-family constant is

$$\rho_*^2 = \lambda_{\max}(G_Y^{\dagger/2} G_U G_Y^{\dagger/2}), \quad (5.3)$$

provided $\text{Ker } G_Y \subseteq \text{Ker } G_U$; otherwise $\rho_* = \infty$. If $\rho_* < 1$ then

$$c^*(G_Y - G_U)c \geq (1 - \rho_*^2)c^*G_Yc. \quad (5.4)$$

This theorem is strictly stronger than checking every endpoint energy. For example, take $G_Y = I_2$ and

$$G_U = \frac{9}{16} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (5.5)$$

Each diagonal of $G_Y - G_U$ is $7/16 > 0$, but its eigenvalues are $-1/8$ and 1 , while $\rho_*^2 = 9/8 > 1$. Both certificates reproduce this mixed-only obstruction exactly.

6. Residual identity and exact finite-risk Feshbach block

For a fixed strict contraction C , define

$$E(h) = \mathcal{U}(h) - C\mathcal{Y}(h), \quad D = I - C^*C, \quad R = I - CC^*. \quad (6.1)$$

Completing the square and using $I + CD^{-1}C^* = R^{-1}$ gives the exact identity

$$\boxed{2\mathcal{A}(h) = \left\| D^{1/2}\mathcal{Y} - D^{-1/2}C^*E \right\|^2 - \left\| R^{-1/2}E \right\|^2.} \quad (6.2)$$

Only the contraction residual is adverse. It cannot be charged again to the source or sextic coordinates already inside $\mathcal{Y}$. It must vanish, factor through a separately reserved owner, or remain in the exact low block.

For an action-signed matrix split into finite danger and complement,

$$H = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}, \quad D \succeq \alpha I > 0, \quad (6.3)$$

congruence gives

$$H \succeq 0 \iff A - BD^{-1}B^* \succeq 0. \quad (6.4)$$

For semidefinite D , retain $\text{Ran } B^* \subseteq \text{Ran } D$ and replace D^{-1} by $D^\dagger$.

The source-null specialization is especially important. In coordinates where

$$\widehat{S}^*\widehat{S} = \begin{pmatrix} 0 & 0 \\ 0 & A_1 \end{pmatrix}, \quad A_1 > 0, \quad H = \begin{pmatrix} H_{00} & H_{01} \\ H_{10} & H_{11} \end{pmatrix}, \quad (6.5)$$

a finite lower source gap requires

$$H_{00} \succeq 0, \quad \text{Ran } H_{01} \subseteq \text{Ran } H_{00}^{1/2}. \quad (6.6)$$

Then

$$H_{\text{eff}} = H_{11} - H_{10}H_{00}^\dagger H_{01}, \quad \mu_* = \lambda_{\min} \left(A_1^{-1/2} H_{\text{eff}} A_1^{-1/2} \right). \quad (6.7)$$

An eigenvector x_* lifts to

$$v_* = \begin{pmatrix} -H_{00}^\dagger H_{01}x_* \\ x_* \end{pmatrix}. \quad (6.8)$$

A negative H_{00} direction is immediately adverse; failure of the range condition forbids any finite source bound; and $\mu_* \leq 0$ supplies an owner-resolved candidate counterdirection. This is the correct matrix test after production inputs exist.

7. Exact family-lock probe split

The A1 family masses and lock give

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}, \quad (7.1)$$

with

$$\mu_1 = 0.0281161723351\dots, \quad \mu_2 = 0.165197152881\dots, \quad \mu_3 = 0.206686674784\dots \quad (7.2)$$

For the scalar kinetic eigenvalue $a(k) \geq 0$,

$$C(k) = (a(k)\mathbf{I} + M)^{-1} = \gamma_0(k)\mathbf{I} + R(k), \quad \gamma_0(k) = \frac{1}{a(k) + \mu_3}, \quad R(k) \succeq 0. \quad (7.3)$$

If $M = V \text{diag}(\mu_i)V^*$, an exact rectangular covariance factor is

$$B(k) = \begin{bmatrix} \sqrt{\gamma_0(k)}\mathbf{I} & R(k)^{1/2} \end{bmatrix}, \quad B(k)B(k)^* = C(k). \quad (7.4)$$

Use independent scalar and anisotropic probes inside (7.4), but use the same probe pair at both endpoints. This preserves every mixed covariance. It is not valid to write $\sqrt{\gamma_0}\mathbf{I} + \bar{R} = \sqrt{\gamma_0}\mathbf{I} + \sqrt{\bar{R}}$ with one probe; that would create a spurious cross covariance.

The synchronous optimal coupling

$$D(k) = C(k)^{1/2} - \sqrt{\gamma_0(k)}\mathbf{I} \quad (7.5)$$

is also exact as a comparison coupling, with

$$\|D(k)\| = \frac{1}{\sqrt{a + \mu_1}} - \frac{1}{\sqrt{a + \mu_3}}. \quad (7.6)$$

But D alone does not carry the whole additive remainder because $R = 2\sqrt{\gamma_0}D + D^2$. Accordingly, the full trace tail below pays R , while the D^2 tail is reported only as a sensitivity diagnostic.

Finally, $R \succeq 0$ is covariance positivity, not action positivity. The entire trace feature $\mathcal{U}$ stays in the negative slot of (4.2). The split separates scales without reversing the owner sign.

8. Analytic derivative-covariance tail

For $L = 16$ write $k = (\pi/8)n$ and $m = \|n\|_\infty$. The A1 scalar kinetic is

$$a(k) = r + Z|k|^2 + Y|k|^4, \quad r - \frac{Z^2}{4Y} = 0.2600000000094749\dots > 0. \quad (8.1)$$

The declared q_0 also gives the safe lower bound

$$a(k) \geq (|k|^2 - q_0^2)^2. \quad (8.2)$$

For all $m \geq N$ put

$$c_N = 1 - \frac{(q_0/h)^2}{N^2}, \quad h = \frac{\pi}{8}.$$

Then

$$a(k) \geq c_N^2 |k|^4. \quad (8.3)$$

The exact sup-shell population is

$$\#\{n : \|n\|_\infty = m\} = (2m + 1)^3 - (2m - 1)^3 = 24m^2 + 2. \quad (8.4)$$

Let $\Delta_\Sigma = (\mu_3 - \mu_1) + (\mu_3 - \mu_2)$ and $\Delta_2 = (\mu_3 - \mu_1)^2 + (\mu_3 - \mu_2)^2$. From the resolvent identity and the mean-value theorem,

$$\sum_i R_i(k) \leq \frac{\Delta_\Sigma}{a(k)^2}, \quad \sum_i D_i(k)^2 \leq \frac{\Delta_2}{4a(k)^3}. \quad (8.5)$$

Using $\zeta(s, N) \leq N^{-s} + N^{1-s}/(s-1)$ gives

$$B_R(N) = \Delta_\Sigma c_N^{-4} h^{-6} \{24\zeta(4, N) + 2\zeta(6, N)\} \quad (8.6)$$

for the derivative trace tail and

$$B_D(N) = \frac{\Delta_2}{4} c_N^{-6} h^{-10} \{24\zeta(8, N) + 2\zeta(10, N)\} \quad (8.7)$$

for the synchronous perturbation's derivative-square tail.

The certificates derive $H_6 = 7083/(500P)$ from R-130 and the two diagnostic headrooms from R-140. With the convention that N is the first included sup-shell, the first values satisfying $H_6 B_R(N)$ below those headrooms are

headroom	N	$H_6 B_R(N-1)$	$H_6 B_R(N)$
0.100434343434376	27	0.1103000379	0.0979877516
0.015912689545308223	49	0.0164887843	0.0154778464

(8.8)

and

$$B_D(27) = 4.10006201 \times 10^{-8}, \quad B_D(49) = 5.64059230 \times 10^{-10}. \quad (8.9)$$

These are explicit progress: the infinite anisotropic high-shell trace can be replaced by a finite low block plus an analytic tail. They are not a production owner estimate because output projections, $C_6(W)$, collar weights, cross reveals, and low couplings have not yet been propagated through (8.6). The infinite-lattice extension is also conditional; A1 itself is certified on finite discrete grids.

9. Production matrix firewall and next finite test

Three layers times the four R-142 active fibre modes give a mandatory 12×12 core only after fixing or aggregating one legal source coordinate per layer and channel. It is not automatically the full matrix. The actual index is schematically

$$i = (\text{reveal}, \text{visit}, q, \text{channel}) \quad \text{plus source-null and returned-low coordinates.} \quad (9.1)$$

Its dimension is not determined by current authority.

For every such coordinate the action-signed entry must contain

$$H_{ij} = \frac{9}{10} G_{ij}^{\text{src}} + \langle \Phi_i, \Phi_j \rangle + \langle \Phi, \Phi_{ij} \rangle - \langle U_i, U_j \rangle - \langle U, U_{ij} \rangle + S_{ij}^{(6)}. \quad (9.2)$$

Every entry must use the same future projection, output cluster, owner partition, and shared trace probe. Primitive trace, future variance, forest, predictable mean, and returned low are multiplicity-audited so that alternate coordinates are never appended as new reserves.

The next decisive computation is therefore:

1. instantiate the corrected coherent $q = 5, 6, 7$ core;
2. augment it by all reveal/visit, phase, source-null, anisotropic-low, and returned-low coordinates;
3. compute $G_Y - G_U$, its kernel condition, and ρ_*^2 ;
4. independently differentiate the complete nonlinear action chart;
5. prove a complement gap, form the exact Feshbach matrix, and return an owner-resolved lifted vector if a nonpositive edge occurs;
6. repeat through $q = 9$ before any $C = 10$ conclusion.

A mixed-Gram pass proves the stronger contraction condition only on the tested feature span. A mixed-Gram failure does not refute T-050 unless its vector is realized as a legal action direction. The independent nonlinear Hessian/Feshbach test supplies that bridge.

10. Separate PDE/BCC route decision

The proposed PDE diagnostics are valuable, but they are not inputs to T-050. They belong to a future Sector-B production-vacuum claim and must not revive the retired B3 claim semantically.

First, $K(q_0) < K(0)$ is shell softness, not linear instability, when both numbers are positive. Second, the current records contain no same-backend restricted surface $F(\phi_{DC}, A_{BCC})$. Third, the historical T-033 calculation tested six separate antipodal-pair restrictions, not the full coupled BCC-star Hessian with cross blocks. Therefore the honest route is

$$\text{discrete shell softness} \longrightarrow \text{commensurate restricted BCC energy} \longrightarrow \text{full coupled BCC-star Hessian.} \quad (10.1)$$

Only after those same-functional tests can the scalar PDE be retained, rejected, or replaced on microscopic grounds. No value from this separate route is used in the R-143 A13 theorem.

11. Evidence-map route summary

- **Old R-142 fixture — superseded as q567, retained as q678:** the physical factor two and sharp shell membership were omitted.
- **Corrected even dyadic family — advanced:** it has exact common output, exact gaps, and one coefficient sign.
- **Diagonal endpoint tests — rejected as sufficient:** positive diagonals do not imply a positive mixed Gram.
- **Complete-feature contraction — advanced conditionally:** it spends source/sextic once and keeps all mixed correlations.
- **PSD covariance as favorable action — rejected:** the trace feature remains in the adverse action slot.
- **Finite low plus analytic tail — advanced conditionally:** the full additive derivative trace is summable with explicit cutoffs.
- **12×12 as the full matrix — rejected:** it is a per-packet core until reveal, visit, and low indices exist.
- **Flat-field evidence as PDE rejection — parked separately:** same-backend restricted energy and a coupled Hessian are missing.

12. Devil's-advocate review

1. **Objection:** The R-142 coherence error invalidates all of R-142.
DISMISSED. The innovation, fibre, covariance, scalar-chart, and band-sign results do not depend on the mislabeled collision. Only the declared q567 coherence evidence is superseded.
2. **Objection:** A passing finite Gram closes the nonlinear Hessian.
UPHELD against that inference. Baseline and second-jet connection terms require the independently differentiated action Hessian.
3. **Objection:** The feature $\mathcal{Y}$ double-spends source or sextic.
DISMISSED. Equation (4.3) checks the exact action coefficients. Forest, mean, and low are not appended as extra reserves.
4. **Objection:** The covariance remainder is PSD, so it helps.
UPHELD against that sign claim. PSD describes covariance. Its trace feature occurs with the adverse minus sign in (4.2).
5. **Objection:** The synchronous D tail pays all of R .
UPHELD. Since $R = 2\sqrt{\gamma_0}D + D^2$, only the additive $R^{1/2}$ probe or the full trace bound pays the complete remainder.
6. **Objection:** The $N = 27, 49$ thresholds are production margins.
UPHELD. Their comparison headrooms are explicitly illustrative, and the owner operations have not been propagated through the tail bound.
7. **Objection:** The corrected coherence proves an adverse action direction.
UPHELD. It proves only coherent coefficient geometry. Complete owner and low/Feshbach blocks can change the sign.
8. **Objection:** The flat numerical field proves that the PDE must be replaced.
UPHELD. A same-functional restricted BCC surface and the coupled star Hessian are prerequisites. The PDE decision is separately registered.

13. Result footer

Result ID: A13-CLASSII-CORRECTED-Q567-FEATURE-CONTRACTION-
COMMON-NOISE-ANISOTROPY-TAIL-BOUNDARY.
Ledger ID: R-143.
Precise statement: corrected dyadic $q=5,6,7$ common-output family;
raw-sign/action-sign factor firewall; exact complete-feature identity;
finite-set Douglas equivalence and residual identity; source-null
Feshbach criterion; exact scalar-plus-anisotropic common-probe covariance
factor; explicit high-shell derivative-trace and synchronous sensitivity
bounds with diagnostic first included shells $N=27$ and $N=49$.
Scope: finite cutoff, positive floor, fixed supplied feature families;
high-shell lattice extension conditional on the registered A1 symbol.
Dependencies: A1-PRODUCTION-FUNCTIONAL-REALISATION; R-125, R-130,
R-140, R-141, R-142.
Evidence grade: ANALYTIC, EXACT, CONDITIONAL, EXECUTED.
Audit status: registered and independently executed under the R-143 manifest.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_corrected_q567_feature_contraction_
common_noise_anisotropy_tail_boundary_verify.py
Expected output: integrated PASS after primary and non-importing independent
certificates, deterministic PDF rebuild, exploration/audit, authority,
manifest, all-page render, and generated-public-surface checks pass.
Falsification gate: any failed physical-output factor, shell membership,
sign conversion, feature coefficient, mixed Gram, kernel/range, covariance
reconstruction, tail threshold, authority/hash, or public-surface check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no production U/Phi matrix or contraction, legal
production adverse vector, owner intertwiner, graph gap, matching, anchor,
A13 gate, OVERLAP_src, Nelson estimate, removal, interacting measure,
Sector-A closure, or T5-T7 claim follows.
Next required action: register the actual reveal/visit/source chart and all
U/Phi base, first-, and second-jet data; assemble the augmented $q=5,6,7$
mixed Gram and independently differentiated action Hessian; then apply the
exact source-null Feshbach test and report an owner-resolved edge vector.